

MIDTERM EXAMINATION
Spring 2009
CS601- Data Communication (Session - 1)

Question No: 1 (Marks: 1) - Please choose one

_____ In RS 422 Balanced mode two lines carry _____ signals which are not identical to each other.

- ▶ same
- ▶ different
- ▶ digital
- ▶ analog

Question No: 2 (Marks: 1) - Please choose one

_____ The _____ generates the data and passes it along with any control information to a _____.

- ▶ DTE; DCE
- ▶ DCE; DTE
- ▶ DCE; node
- ▶ DTE; application

Question No: 3 (Marks: 1) - Please choose one

_____ In _____ transmission, a start bit and a stop bit frame a character byte.

- ▶ Asynchronous serial
- ▶ Synchronous serial
- ▶ Parallel
- ▶ Asynchronous & Synchronous serial

Question No: 4 (Marks: 1) - Please choose one

_____ In _____ transmission, bits are transmitted simultaneously, each across its own wire.

- ▶ Asynchronous serial
- ▶ Synchronous serial
- ▶ Parallel
- ▶ Asynchronous & Synchronous serial

Question No: 5 (Marks: 1) - Please choose one

Asynchronous transmission is _____

- ▶ slow
- ▶ costly
- ▶ non-effective
- ▶ fast

Question No: 6 (Marks: 1) - Please choose one

_____ In
_____ transmission the timing of the signal is unimportant.

- ☐ Asynchronous
- ☐ Synchronous
- ☐ Polar
- ☐ Bi-polar

Question No: 7 (Marks: 1) - Please choose one

_____ In
8QAM each signal shift or one baud represents _____.

- ☐ 4 bits
- ☐ 2 bits
- ☐ 5 bits
- ☐ 3 bits

Question No: 8 (Marks: 1) - Please choose one

_____ PCM is the first process of PAM.

- ☐ True
- ☐ False

Question No: 9 (Marks: 1) - Please choose one

_____ Bi
phase encoding is a type of bipolar encoding in which we use two voltage levels.

- ☐ True
- ☐ False

Question No: 10 (Marks: 1) - Please choose one

_____ The
amplitude of a digital signal depends upon the _____ to represent a bit.

- ☐ phase
- ☐ voltage
- ☐ wavelength

Question No: 11 (Marks: 1) - Please choose one

_____ Time domain plot show changes in signal phase with respect to time.

- ☐ True
- ☐ False

Question No: 12 (Marks: 1) - Please choose one

_____ OSI
model was developed prior to TCP/IP model.

- ☐ True
- ☐ False

Question No: 13 (Marks: 1) - Please choose one

_____ The
_____ layer is the layer closest to the transmission medium.

- ▶ physical
- ▶ data link
- ▶ network
- ▶ transport

Question No: 14 (Marks: 1) - Please choose one

_____ Data
chunk at data link layer is called_____

- . ▶ frame
- . ▶ packet
- . ▶ datagram

Question No: 15 (Marks: 1) - Please choose one

_____ The
internet model consists of _____ layers.

- ▶ three
- ▶ two
- ▶ five
- ▶ seven

Question No: 16 (Marks: 1) - Please choose one

_____ representation of links that connect nodes is called as physical topology.

- . ▶ geometrical
- . ▶ logical
- . ▶ physical

Question No: 17 (Marks: 1) - Please choose one

_____ is a multipoint topology.

- ▶ Ring
- ▶ Mesh
- ▶ Tree
- ▶ Bus

Question No: 18 (Marks: 1) - Please choose one

_____ In
mesh topology, if there are five nodes then there will be _____ links.

- ▶ 5
- ▶ 10

▶ 15

▶ 20

Question No: 19 (Marks: 1) - Please choose one

proprietary standard is also termed as de facto standard.

▶ True

▶ False

Question No: 20 (Marks: 1) - Please choose one

Effective network mean that the network has fast delivery, timeliness and

▶ high bandwidth

▶ duplex transmission

▶ accurate transmission

▶ low bandwidth

Question No: 21 (Marks: 2)

What is purpose of distributed processing?

Question No: 22 (Marks: 3)

What is HDB3.

Question No: 23 (Marks: 5)

Differentiates between the following terms. [10 marks]

(a) Syntax and Semantics of protocol.

(b) Network management and exchange management.

(c) Monolithic and structured.

Question No: 24 (Marks: 10)

Differentiate between the following terms.

a) Frequency spectrum and bandwidth.

(b) Bandwidth and throughput.

(c) Bit interval and bit rate.

Non